Assessment of academic achievement: Project work

The project work consists of carrying out a data analysis for your own data. You document this data analysis in a portfolio that you create during the semester. The data analysis will consist of three parts:

1. **explorative data analysis**
2. **classification**
3. **regression**

For each part you have to hand in:

- **Jupyter Notebook**
- **Documentation**
- **Feedback an fellow students work**
- **Reflection sheet**

General requirements are:

The Jupyter Notebook

- can be executed without errors;
- is clearly structured;
- is explained in Markdown sections;
- is clearly documented;
- contains no superfluous code.

The documentation (max 3 pages)

- is written and structured in a scientific style.
- presents and explains the results of the Python analyses.
- includes a clear and concise description of the methodology used to analyse the data;
- discusses the relevance of the results in the context of the problem under investigation and draws conclusions based on the analyses.

Data Science/
Barbara Priwitzer

The feedback

- is problem-related and specific;
- shows understanding for the procedure;
- gives helpful advice;
- is respectful and critical.

The reflection

- is based on a form that you are required to use.
- The main purpose of reflection is to present your ideas on how to carry out a data analysis. Among other things, this is to ensure that you do not simply follow suggestions from generative AI without questioning them.
- According to the form you should
    - describe two different expectations you had of the data analysis;
    - discuss two different problems you encountered during the data analysis and how you solved them;
    - explain how you could improve your work with the feedback from a fellow student.

Using generative AIs

Copilot

All university members can use Bing Copilot free of charge via their universityMicrosoft account.

Bing Copilot always has access to GPT-4 or, outside peak times (times when demand for services and resources is at its highest), to GPT-4 Turbo. The service is encrypted and no data is passed on to GPT (Generative Pre-trained Transformer). Data entered remains internal and is deleted after the browser is closed.

You also receive citations including sources, which is of course a great advantage for scientific work.

When logging in, select 'Log in with a business, school or university account'. Use your university Microsoft account here (university email address and associated password including two-factor authentication).

Important!

No personal data, especially about third parties, and no university-related information may be entered into the AI system.

The data from Microsoft Bing Copilot will not be used to train the algorithms. Prompts, responses and data accessed via Microsoft Graph will not be used for training Foundation LLMs, including those used by Microsoft 365 Copilot.

GitHub Copilot

GitHub Copilot in Visual Studio Code

GitHub Copilot is an AI coding assistant that helps you write code faster and with less effort, allowing you to focus more energy on problem solving and collaboration.

You can use Copilot in your IDE to:

- Get code suggestions as you type in your IDE
- Chat with Copilot to ask for help with your code
- Ask Copilot for help using the command line

To use Copilot in VS Code, you need access to a GitHub Copilot subscription. Verified students, teachers, or open source maintainers may be eligible to use Copilot for free.

You can set up GitHub Copilot in VSC:
https://code.visualstudio.com/docs/copilot/setup#_set-up-copilot-in-vs-code

For the usage of GitHub Copilot consult the following documentation:
https://docs.github.com/en/copilot

Be aware

As you all know, generative AIs can hallucinate, i.e. present false information in a very convincing manner. You must therefore always make sure:

- Is the information correct? You must refer to the theory, specifications and requirements from the Data Science module.
- Statements made by generative AI are often generally correct, or at least not incorrect, but extremely general and unspecific. For example: ‘Regression is a powerful tool for understanding relationships, quantifying them and making predictions.’ Be sure to avoid using such statements without revising them.
- Also check the code: It is usually executable without errors, but generative AIs often produce overly complex variants or use too many advanced functions. In some cases, data or features are also reinterpreted in an inadmissible manner in order to be able to apply certain methods. Always adhere to the specifications of the Data Science module. The aim is to use methods that you have learned and that you understand, rather than applying new functions whose background you are unfamiliar with.
- Urge the AI to explain what it has done and why. Use the AI more for debugging than for coding.

Part 1: Exploratory Data Analysis

The following aspects are assessed for the exploratory data analysis:

- What do the data describe? Where do they come from?
- How many data sets are there? How many features? How are the features scaled?
- What is interesting about the data? What questions can be asked?
- Are values missing? How many? Why are they missing? Can they be meaningfully replaced?
- How can important features be visualised? (at least two graphs, different types) What can be said about the data with the help of the graphs? Are the graphical representations complete, appropriate and meaningful? Are they interpreted correctly?
- What are the values of location and dispersion measures for important features? What information about the data can be derived from these values? (here it can be useful to group the data and e.g. compare mean values of a feature for different groups) Are the calculated measures of position and dispersion presented clearly and informatively? Do they allow helpful conclusions 4ft h drawn about the data?
- Are there outliers in important features? How can this be recognised? Are there many outliers? Where could they come from? How could the outliers be taken into account? What information about the data does this provide?
- For the interpretation of the results from the exploratory data analysis in the documentation correct, helpful and informative?

Part 2: Classification

The following aspects are evaluated for the classification:

- Which discrete or qualitative features (classes) should be analysed? It may be possible to divide a continuous feature into classes. Formulate meaningful classification tasks. Explain which features you use as independent variables for the classification and why.
- Carry out at least two different classifications: different classification tasks, different classifiers, different parameters. Give your reasoning for the choice of classifiers. Pay attention to the correct execution of the classifications in Python.
- If possible, visualise the models and the classification results graphically. Give an interpretation of the graphical visualisation.

- Use meaningful training scenarios and explain why you chose them. Use the division into training, validation and test set correctly or carry out cross-validation.
- Evaluate the models with suitable measures and compare the classification models. Investigate which features contribute significantly to the classification. Give a meaningful interpretation of these measures.
- Make predictions for new data using the models. How would you rate the classification result?
- Is the interpretation of the results from the classifications in the documentation correct, helpful and informative?

Part 3: Regression

The following aspects are evaluated for the regression:

- Which continuous features are interesting to be analysed? Which correlations are interesting?
- Are the features correlated? How high is the correlation? Can you exclude spurious correlation?
- Examine different regression models: linear regression, generalised linear regression, multiple linear regression. Make sure the regressions (at least two different ones) have been carried out correctly. Give a reasoining for the choice of regression models.
- Compare the models with suitable criteria. Which one is particularly good according to your analysis?
- Use the models to make predictions for new data. Do the predictions make sense?
- If possible, visualise the models graphically. The graphical representations for the regressions are complete, correct and informative. Give an interpretation of the graphical visualisation.
- Is the interpretation of the results from the regression analysis in the documentation correct, helpful and informative?

Assessment

You can achieve the following points for every part (30 pts):

- Jupyter Note book: 8 pts
- Documentation: 9 pts
- Feedback: 4 pts
- Reflection: 5 pts
- Form: 4 pts

Documentation/ Jupyter Notebook: Tasks marked in green must be completed – otherwise points will be deducted; tasks marked in yellow should be completed if possible.

Form includes: Structure, style, grammar, spelling, references; structure of Jupyter Notebooks, annotation, use of own Python functions

This results in a maximum of 90 points for the project work.

The following grades are given for Data Science:

Points	81 – 90	77 – 80	73 – 76	69 – 72	65 – 68	61 – 64	57 – 60	53 – 56	49 – 52	45 - 48	< 45
Marks	1,0	1,3	1,7	2,0	2,3	2,7	3,0	3,3	3,7	4,0	Failed

Use of generative AI

You may use generative AI for code creation or code correction, such as ChatGPT or Copilot. Please indicate this use in the sources according to the following template:

AI tool	**Type of use**	**Affected part**	**Remarks**
Copilot	code generation	Exploratory data analysis: Creation of graphs	Used only for more appropriate selection of colours
ChatGPT	Code - Generation	classification: Use of decision tree	Code was compared with the template and adapted
Gemini	Text revision	Summarising the results	Revising and supplementing a draft text